KAIZEN MATHS

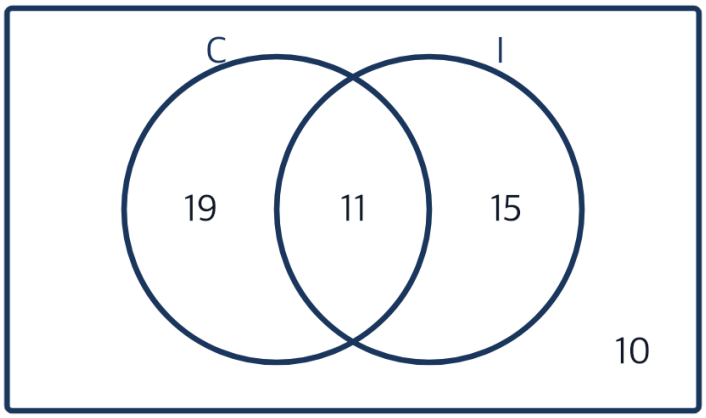
Lesson starter

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

SA1



55 people at a party were surveyed. 30 chose Cake, 26 chose Ice cream, and 11 chose both. Find the number who chose **only Cake**.

STA-B8CE-L2 · SA

Show your working

KAIZEN MATHS

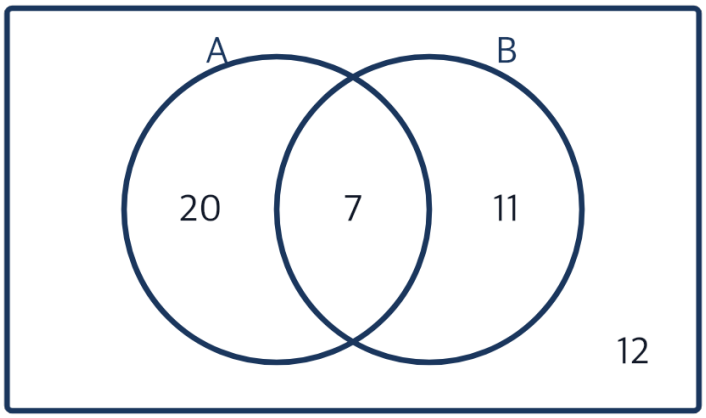
Lesson starter

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

SB1



Use the diagram to find **P(A | B)**.

STA-B8CE-L2 · SB

Show your working

KAIZEN MATHS

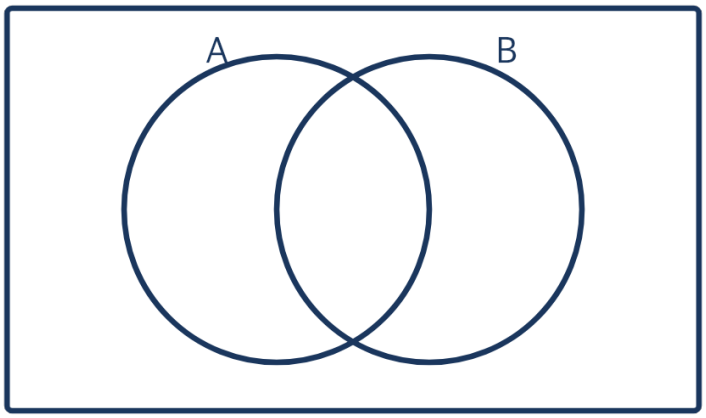
Lesson starter

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

SC1



Given n(A)=21, n(B)=29, and n(A ∪ B)=46, find n(A ∩ B).

STA-B8CE-L2 · SC

Show your working

KAIZEN MATHS

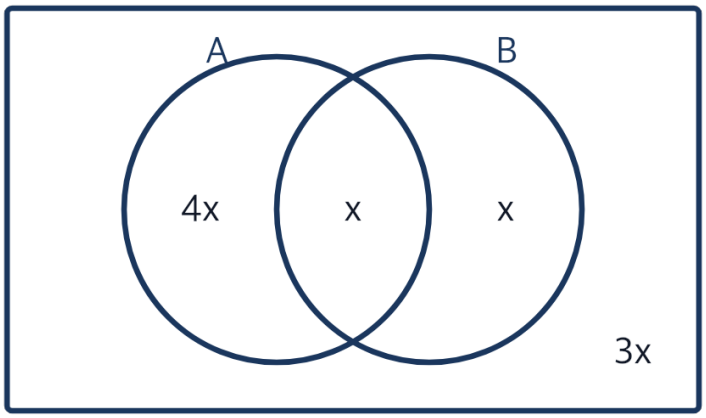
Lesson starter

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

SD1



The diagram shows probabilities in each region. Find **x**, then find **P(A′ ∩ B)**.

STA-B8CE-L2 · SD

Show your working

KAIZEN MATHS

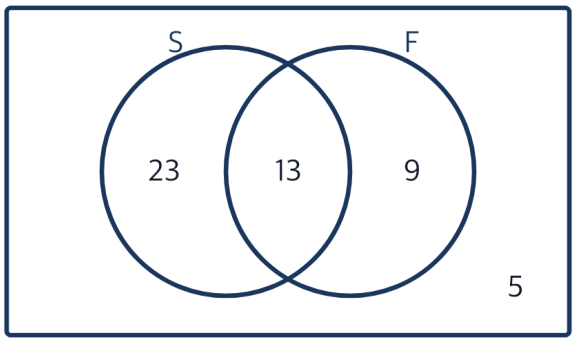
Exit ticket

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

EA1



50 students were surveyed. 36 chose Spanish, 22 chose French, and 13 chose both. Find the number who chose **only Spanish**.

STA-B8CE-L2 · EA

Show your working

KAIZEN MATHS

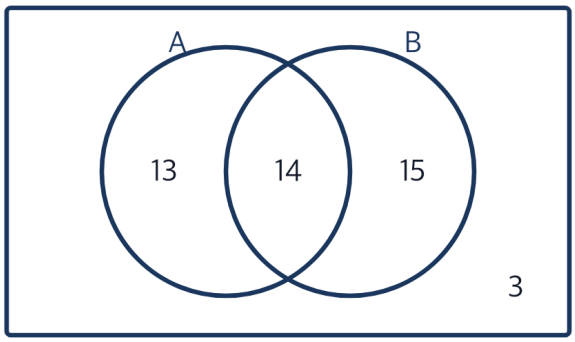
Exit ticket

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

EB1



Use the diagram to find **P(A ∩ B)**.

STA-B8CE-L2 · EB

Show your working

KAIZEN MATHS

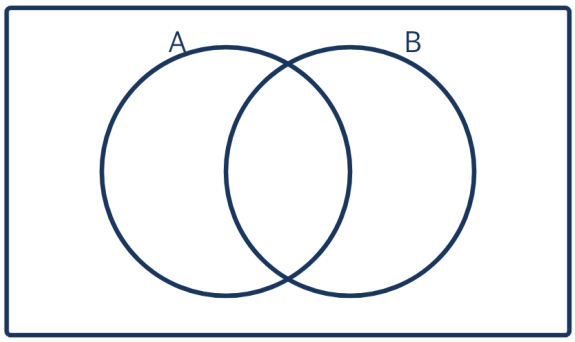
Exit ticket

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

EC1



Given n(A)=37, n(B)=35, and n(A ∪ B)=56, find n(A ∩ B).

STA-B8CE-L2 · EC

Show your working

KAIZEN MATHS

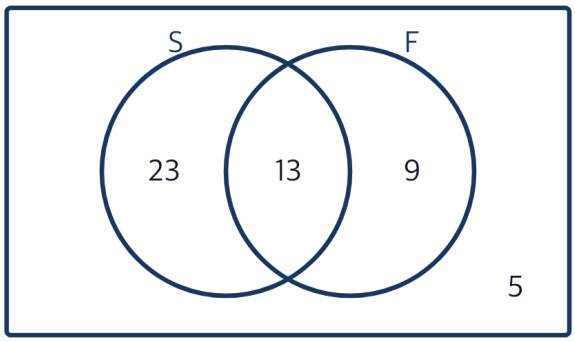
Exit ticket

Venn Diagrams and Set Probability

Two-Set Applications

Name _____

EA1



50 students were surveyed. 36 chose Spanish, 22 chose French, and 13 chose both. Find the number who chose **only Spanish**.

STA-B8CE-L2 · EA

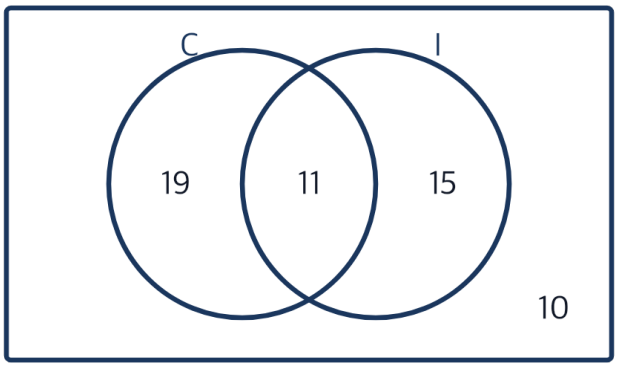
Show your working

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Applications

STA-B8CE-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



55 people at a party were surveyed. 30 chose Cake, 26 chose Ice cream, and 11 chose both. Find the number who chose **only Cake**.

Answer:

19

Working:

Start in the overlap: n(C ∩ I) = 11.

$n(C \cap I') = n(C) - n(C \cap I) = 30 - 11 = 19.$

This is Cake only.

$n(C' \cap I) = n(I) - n(C \cap I) = 26 - 11 = 15.$

This is Ice cream only.

$n(C \cap I') = 30 - 11 = 19.$

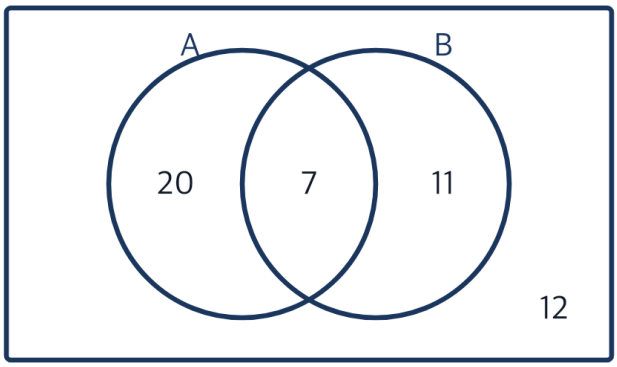
This is Cake only: chose Cake but not Ice cream.

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Applications

STA-B8CE-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Use the diagram to find **P(A | B)**.

Answer:

$\frac{7}{18}$

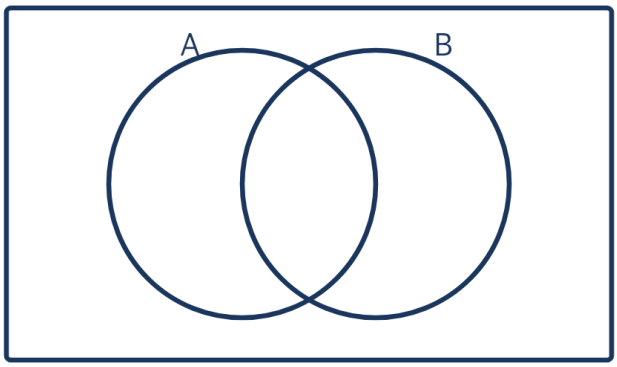
Working:

For probability, write the required number over the correct total.

$P(A \mid B) = \frac{7}{18}$

Simplify to $\frac{7}{18}$.

SC1



Given n(A)=21, n(B)=29, and n(A ∪ B)=46, find n(A ∩ B).

Answer:

4

Working:

Use $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

$46 = 21 + 29 - n(A \cap B)$.

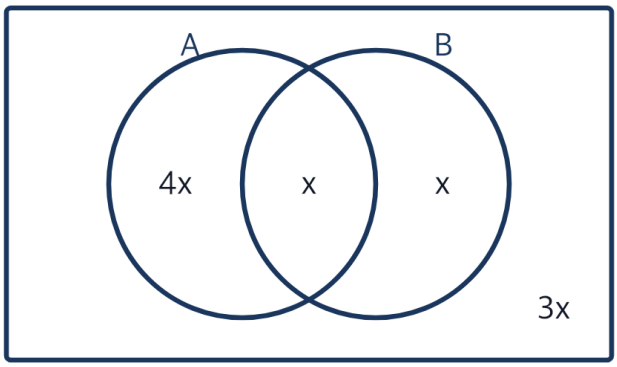
$n(A \cap B) = 50 - 46 = 4$.

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Applications

STA-B8CE-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



The diagram shows probabilities in each region. Find **x**, then find **P(A' ∩ B)**.

Answer:

$\frac{1}{9}$

Working:

All probabilities in the universal set add to 1.

$4x + x + x + 3x = 1$

$9x = 1$

$x = \frac{1}{9}$

Substitute x = $\frac{1}{9}$ into each region.

$P(A' \cap B)$

This is the B-only region.

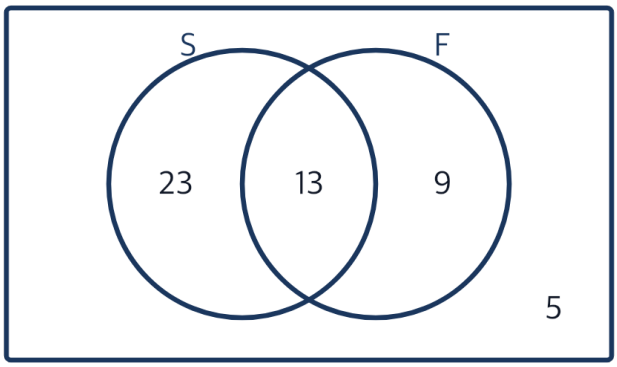
$P(A' \cap B) = \frac{1}{9} = \frac{1}{9}$.

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Applications

STA-B8CE-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



50 students were surveyed. 36 chose Spanish, 22 chose French, and 13 chose both. Find the number who chose **only Spanish**.

Answer:

23

Working:

Start in the overlap: n(S ∩ F) = 13.

$n(S \cap F') = n(S) - n(S \cap F) = 36 - 13 = 23.$

This is Spanish only.

$n(S' \cap F) = n(F) - n(S \cap F) = 22 - 13 = 9.$

This is French only.

$n(S \cap F') = 36 - 13 = 23.$

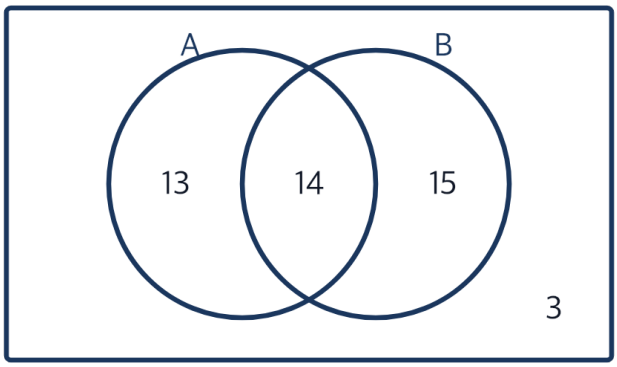
This is Spanish only: chose Spanish but not French.

Teacher answers & working

Venn Diagrams and Set Probability · Two-Set Applications

STA-B8CE-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Use the diagram to find **P(A ∩ B)**.

Answer:

$\frac{14}{45}$

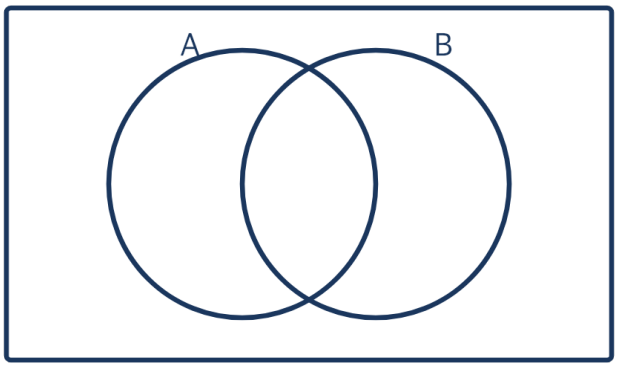
Working:

For probability, write the required number over the correct total.

$P(A \cap B) = \frac{14}{45}$

Simplify to $\frac{14}{45}$.

EC1



Given n(A)=37, n(B)=35, and n(A ∪ B)=56, find n(A ∩ B).

Answer:

16

Working:

Use $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

$56 = 37 + 35 - n(A \cap B)$.

$n(A \cap B) = 72 - 56 = 16$.